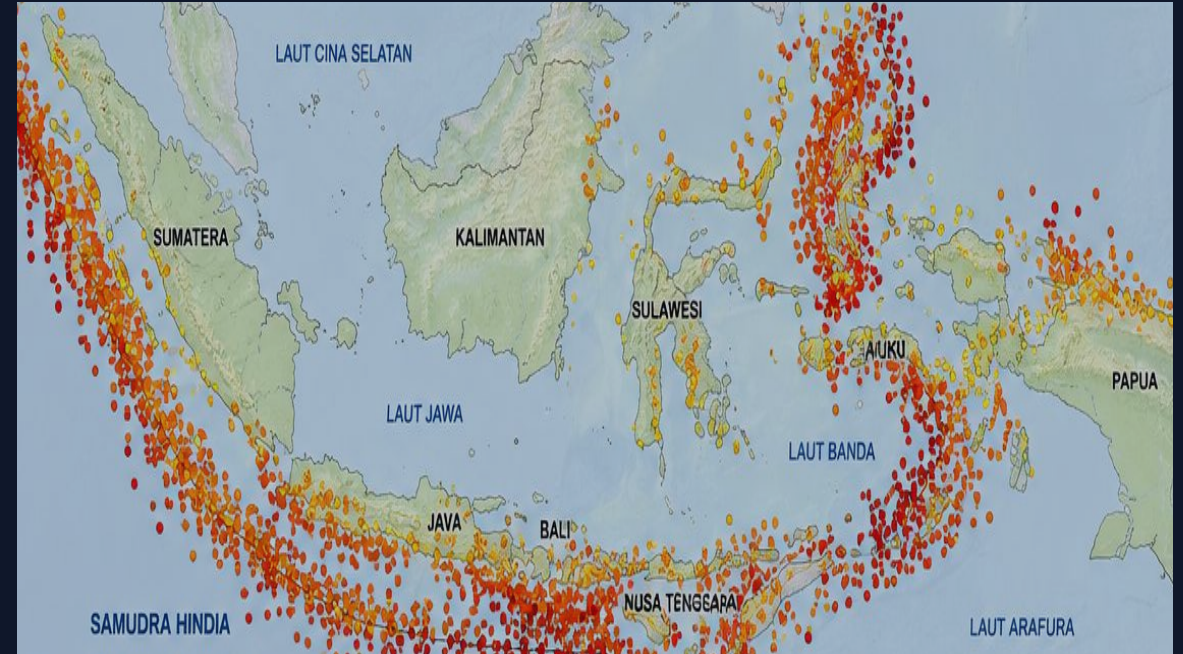


Earthquake Pattern Discovery using Association Rule Mining

Discovering Hidden Spatial–Temporal
Relationships from Historical Indonesian
Earthquake Data.

Silvia Sholikatul Khasanah

🔗 Python • Pandas • Association Rule Mining • Apriori



Why This Project?



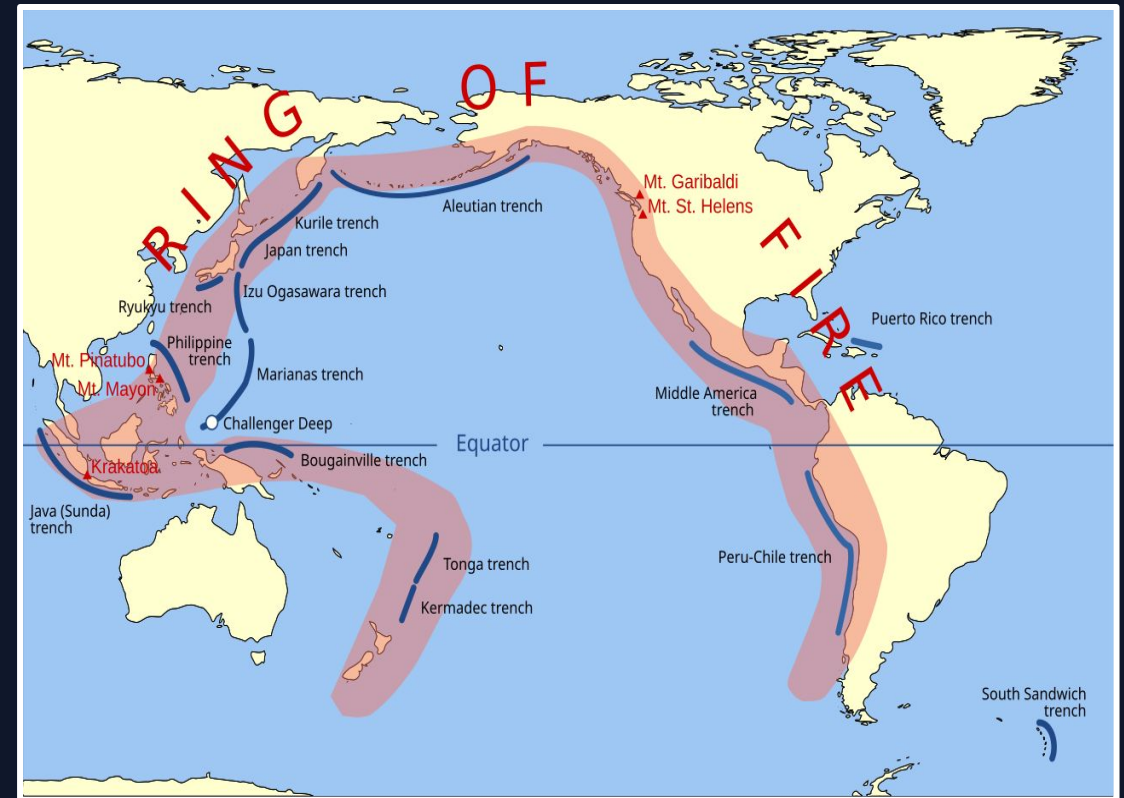
High Seismic Activity: Indonesia is globally recognized as one of the countries with the highest seismic activity due to its location on the Pacific Ring of Fire.



Massive Historical Data: The BMKG dataset has recorded over **91,395. earthquake events** spanning a 13-year period.



The Missing Link: Existing data primarily describes **individual events**, lacking insights into the hidden spatial relationships and co-occurrences between different regions.



Research Questions



Co-Occurrence

Which specific earthquake regions in Indonesia frequently experience seismic activity together?



Rule Strength

How strong are these spatial associations, and can we quantify their statistical significance?



Temporal Patterns

Are there recurring spatial-temporal patterns present across the historical observation period?

Raw Events



Hidden Relationships



Knowledge Discovery

Dataset Overview

	date	time	latitude	longitude	depth	magnitude	remark
0	01/01/2009	18:37:24.148	-4.37	101.12	10	5.1	Southern Sumatra - Indonesia
1	01/01/2009	16:57:53.325	-4.42	101.07	10	5.3	Southern Sumatra - Indonesia
2	01/01/2009	10:35:34.583	1.42	121.99	10	5.0	Minahassa Peninsula - Sulawesi
3	01/01/2009	08:42:12.410	1.47	122.12	10	4.5	Minahassa Peninsula - Sulawesi
4	01/01/2009	05:27:19.531	0.77	121.22	10	4.9	Minahassa Peninsula - Sulawesi

Metric	Value / Detail	Description
Total Events	91,395	Individual earthquake occurrences recorded by BMKG.
Observation Period	2009 – 2022	14 years of continuous seismic monitoring data.
Unique Regions	51	Distinct geographical zones identified in the catalog.
Daily Earthquake Transactions	4,333	Total days with at least one earthquake event (Transaction baseline).

NO.	Unique Regions
1	Southern Sumatra - Indonesia
2	Minahassa Peninsula - Sulawesi
3	Banda Sea
4	Bali Sea
5	Sulawesi - Indonesia
6	Ceram Sea
7	Irian Jaya Region - Indonesia
8	Borneo
9	Southern Molucca Sea
10	Flores Region - Indonesia
11	Halmahera - Indonesia
12	Talaud Islands - Indonesia
13	Celebes Sea
14	Java - Indonesia
15	Flores Sea
16	Aru Islands Region - Indonesia
17	Timor Region
18	Northern Sumatra - Indonesia
19	Off West Coast of Northern Sumatra
20	Sunda Strait - Indonesia
21	Tanimbar Islands Reg. - Indonesia
22	Sumbawa Region - Indonesia
23	Java Sea
24	South China Sea
25	Northern Molucca Sea

NO.	Unique Regions
26	South of Bali - Indonesia
27	South of Java - Indonesia
28	Near North Coast of Irian Jaya
29	Mindanao - Philippines
30	Seram - Indonesia
31	Buru - Indonesia
32	Southwest of Sumatra - Indonesia
33	Irian Jaya - Indonesia
34	Bali Region - Indonesia
35	Timor Sea
36	North of Halmahera - Indonesia
37	Sulu Archipelago - Philippines
38	Philippine Islands Region
39	Malay Peninsula
40	Sumba Region - Indonesia
41	Near N. Coast of New Guinea - PNG.
42	South of Sumbawa - Indonesia
43	W. Caroline Islands - Micronesia
44	East of Philippine Islands
45	Savu Sea
46	South Indian Ocean
47	Arafura Sea
48	Near S. Coast of New Guinea - PNG.
49	New Guinea - Papua New Guinea
50	Ninigo Islands Region - P.N.G.
51	Near South Coast of Irian Jaya

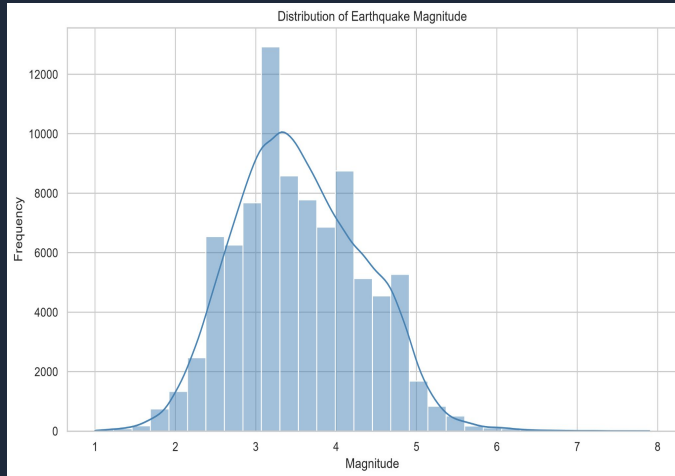
Dataset Quality Assessment

-  **No Missing Values:** The dataset is structurally complete with zero null values across all critical features.
-  **Deduplication:** All duplicate records resulting from sensor overlaps have been rigorously removed.
-  **Standardized Regions:** Mapped effectively into **51 unique geographic regions**.
-  **Average Magnitude:** 3.55 Mw, with the distribution concentrated below magnitude 4.0.
-  **Average Depth:** Recorded at 48 km (predominantly shallow earthquakes)

Metric	Value
Rows	91395
Columns	7
Missing Values	0
Duplicate Rows	2687

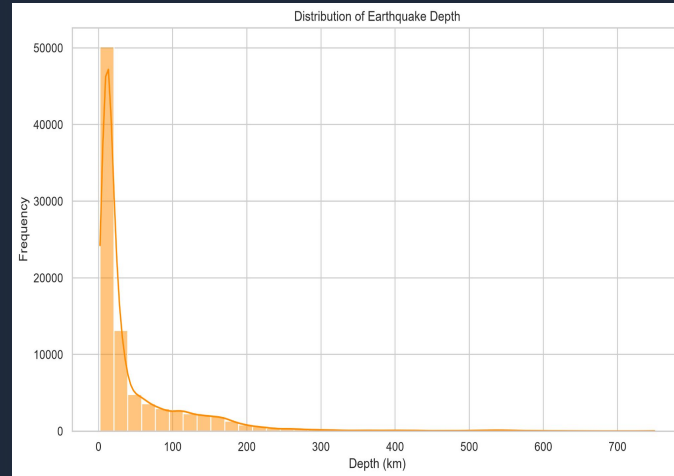
	count	mean	std	min	25%	50%	75%	max
latitude	88708.0	-3.429591	4.348787	-11.00	-7.93	-2.92	0.11	6.0
longitude	88708.0	119.127935	10.747156	94.02	113.49	121.09	126.84	142.0
depth	88708.0	48.328257	75.703764	2.00	10.00	15.00	52.00	750.0
magnitude	88708.0	3.549652	0.800637	1.00	3.00	3.50	4.10	7.9

Exploratory Data Analysis



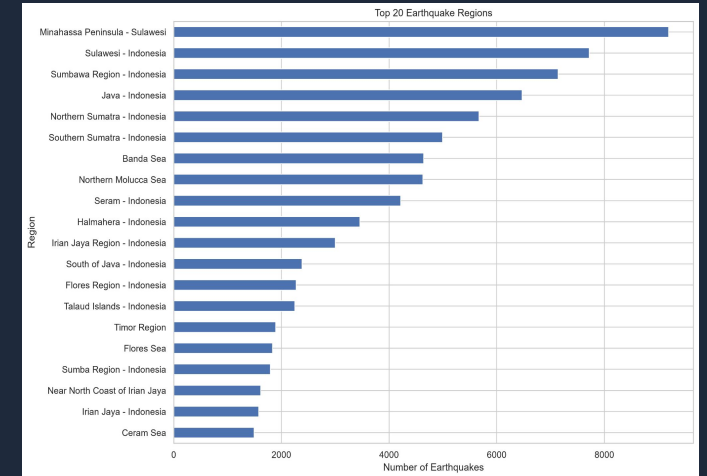
Magnitude

Most earthquakes are concentrated between magnitude 3 and 4.



Depth

Approximately 66% of events are classified as shallow earthquakes (0–30 km).

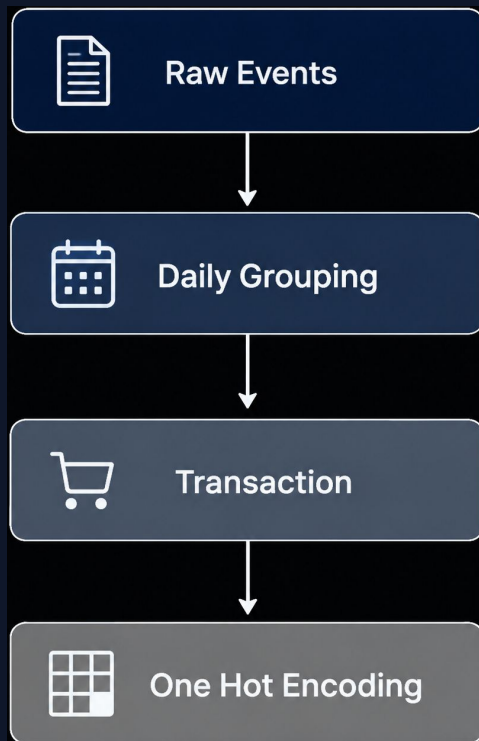


Top Regions

The Minahassa Peninsula emerges as the most frequently occurring region across the entire 13-year period.

From Earthquake Events to Transactions

Because the Apriori algorithm requires transactional data, individual earthquake events are grouped into **daily transactions**.



1. Raw Events

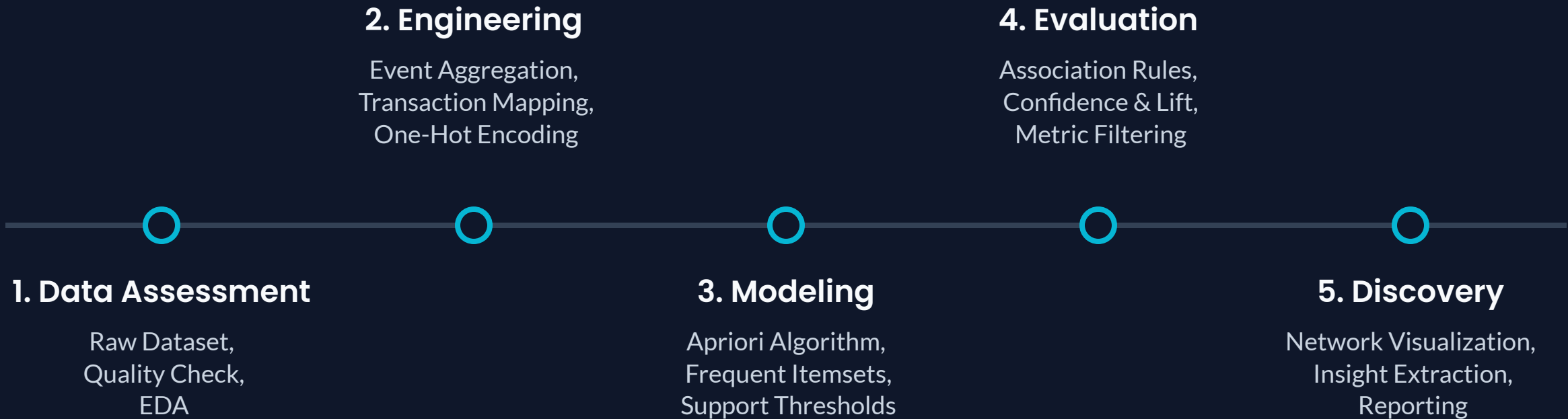
Date	Region
1 Jan	Java
1 Jan	Banda Sea
1 Jan	Sulawesi

2. Transaction Encoding

Transaction T1:
{ "Java", "Banda Sea", "Sulawesi" }

TxID	Java	Banda Sea	Sulawesi
T1	1	1	1

Data Science Methodology



Frequent Itemsets Discovery

Algorithm Parameters

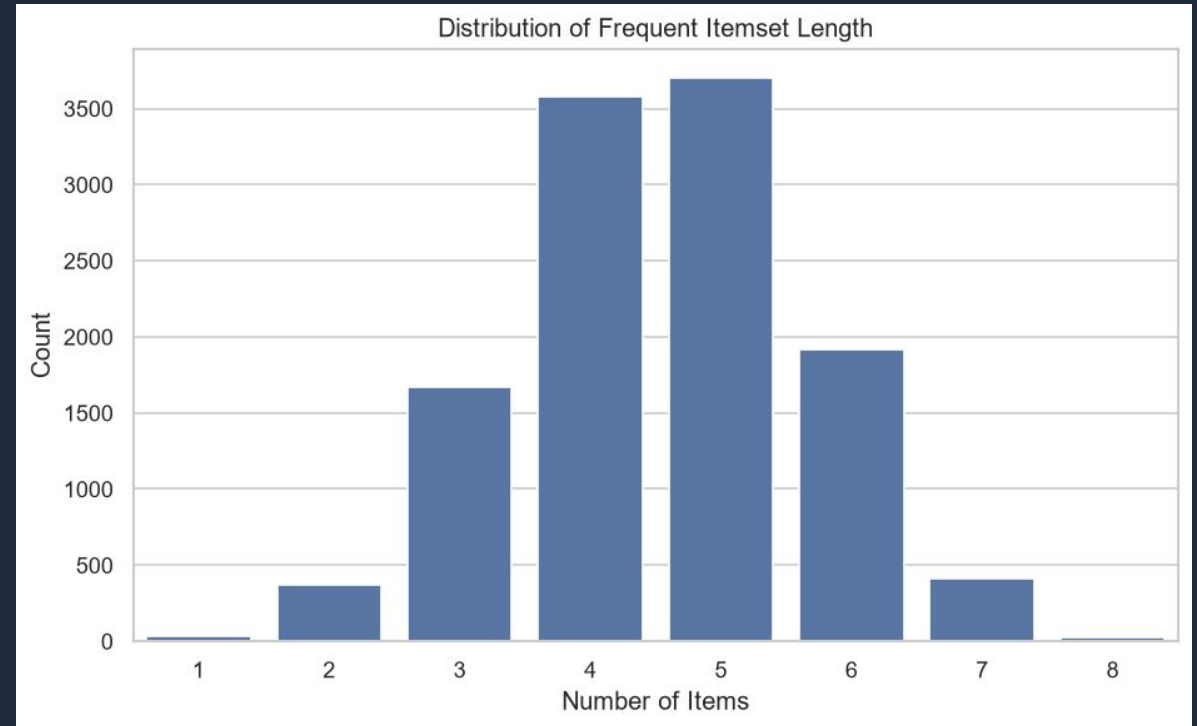
MINIMUM SUPPORT

4.0%

RESULTING FREQUENT ITEMSETS

11,708

Note : selected after sensitivity analysis



Top Discovered Regions

#1 Minahassa

#2 Java

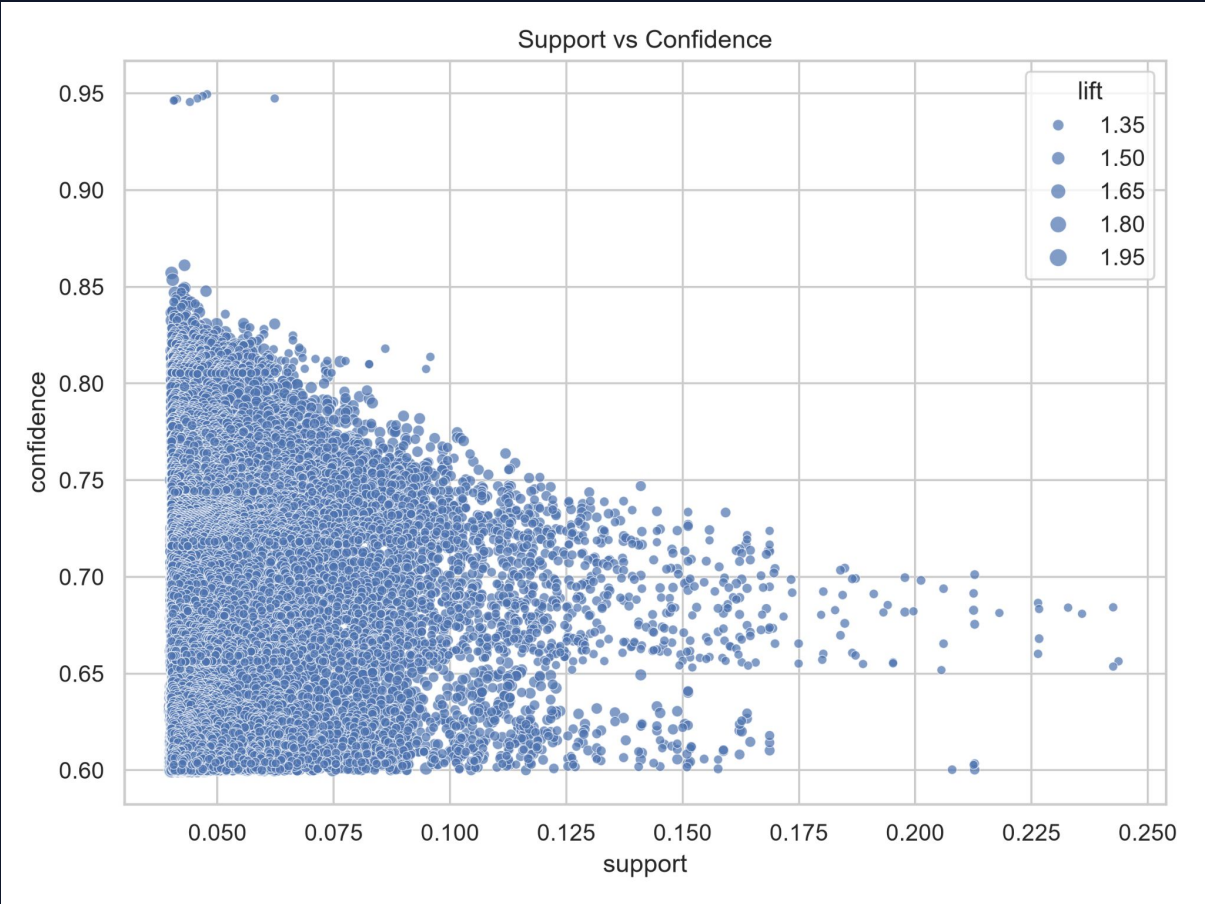
#3 Banda Sea

Association Rule Evaluation

Association rules explain the directional relationships between regions using three primary metrics:

Support, Confidence, and Lift.

Metric	Value
Total Rules Generated	27,129
Average Confidence	69.8%
Average Lift	1.33
Maximum Lift	1.97

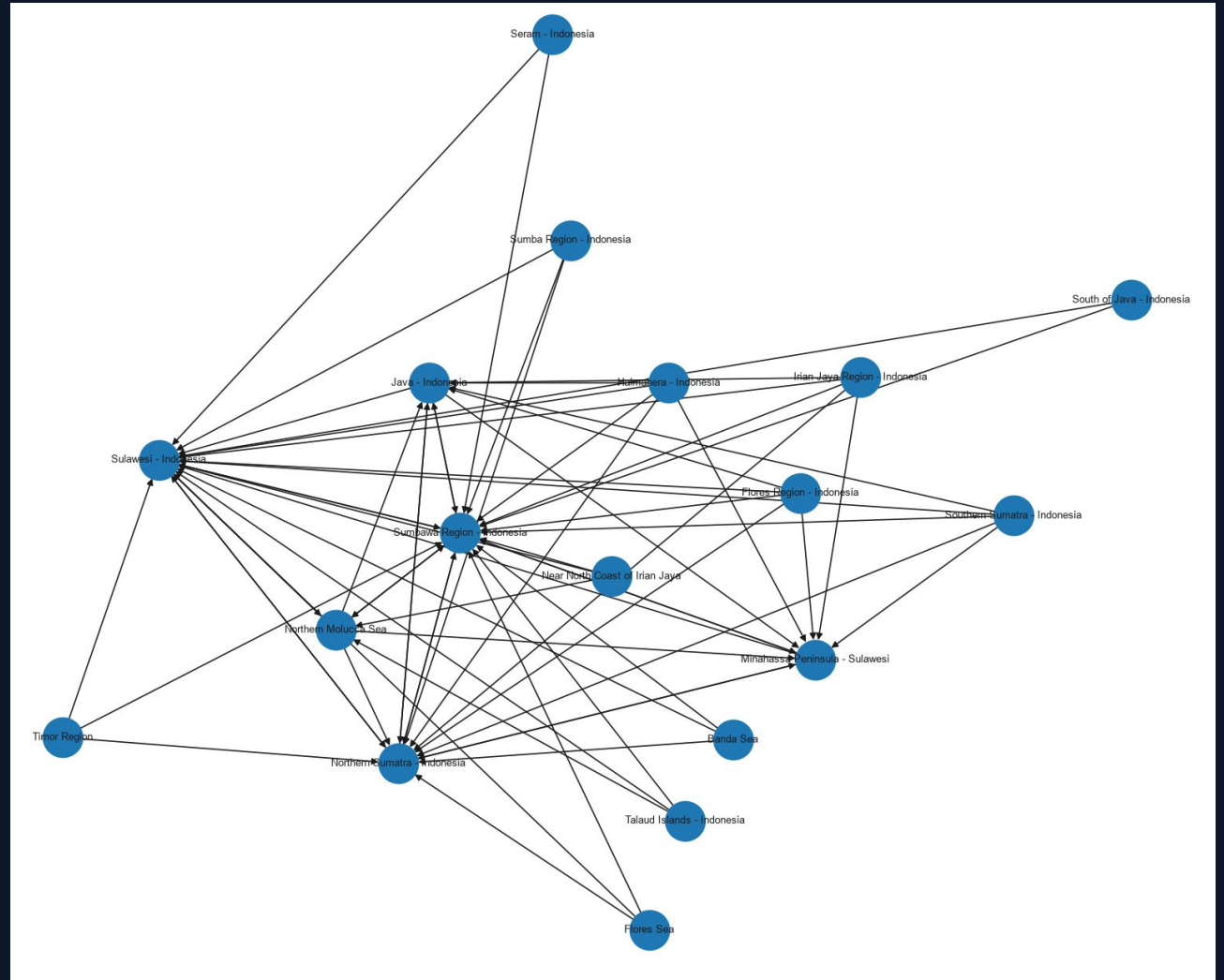


Association Network

Visualizing the spatial relationship ecosystem.

Most Connected Regions

- Minahassa Peninsula
- Java
- Banda Sea
- Northern Sumatra
- Halmahera



Project Results Dashboard

91,395

Earthquakes

4,333

Transactions

51

Regions

11,708

Frequent Itemsets

27,129

Association Rules

69.8%

Average Confidence

Key Insights

Insight 1: Core Hotspot

The **Minahassa Peninsula & Sulawesi** region acts as the primary anchor, exhibiting the highest frequency of occurrences and acting as a central node in the association network.

Insight 2: Dense Topology

A staggering **27,129 significant association rules** were discovered, proving that seismic activity across Indonesian earthquake regions exhibit strong spatial co-occurrence patterns within daily observations.

Insight 3: High Dependency

The strongest rules yielded a **Lift of 1.97**, indicating that these co-occurrences are nearly twice as likely to happen together compared to purely random chance.

Insight 4: Algorithmic Value

Association Rule Mining effectively uncovers hidden spatial relationships that are virtually impossible to detect through manual inspection of raw historical data.

Conclusion

- ✓ Successfully transformed 91,395 raw earthquake events into 4,333 actionable daily transactions.
- ✓ Generated 11,708 frequent itemsets identifying core regional activity zones.
- ✓ Produced 27,129 association rules detailing spatial co-occurrences.
- ✓ Demonstrated a complete, end-to-end unsupervised learning data science workflow.
- ✓ Revealed meaningful, statistically significant spatial-temporal earthquake patterns.

Limitations & Future Work

Current Limitations

- ✗ Strictly pattern discovery; provides no causal inference.
- ✗ Assumes a static 24-hour (daily) observation window.
- ✗ Relies entirely on historical data, lacking predictive modeling.

Future Work

- Compare performance with FP-Growth algorithms.
- Implement Sequential Pattern Mining for temporal tracking.
- Adjust temporal windows (hourly, weekly) for sensitivity analysis.
- Integrate pipeline with real-time earthquake telemetry feeds.

Thank You

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